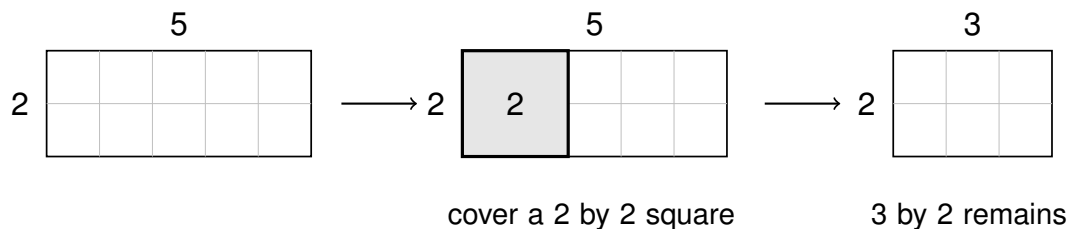
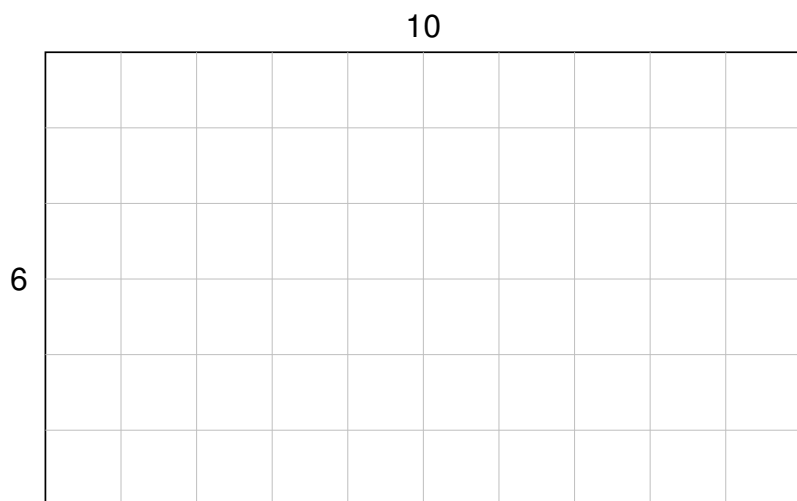
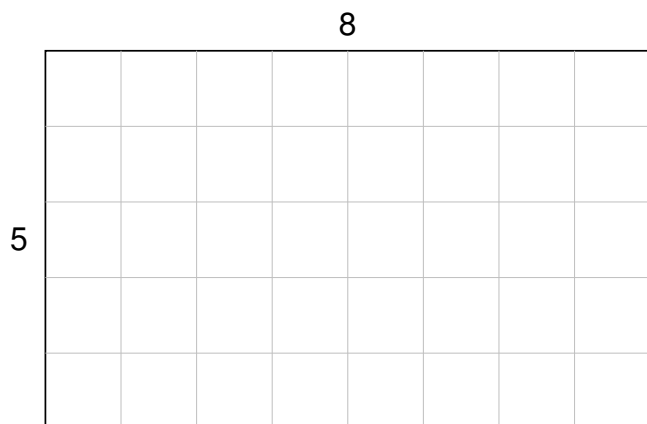


Keep square edges on grid lines. Cover rectangles with no gaps, overlaps, or overhang. For the biggest-square rule, cover the largest square at one end, then repeat on the rectangle left over.

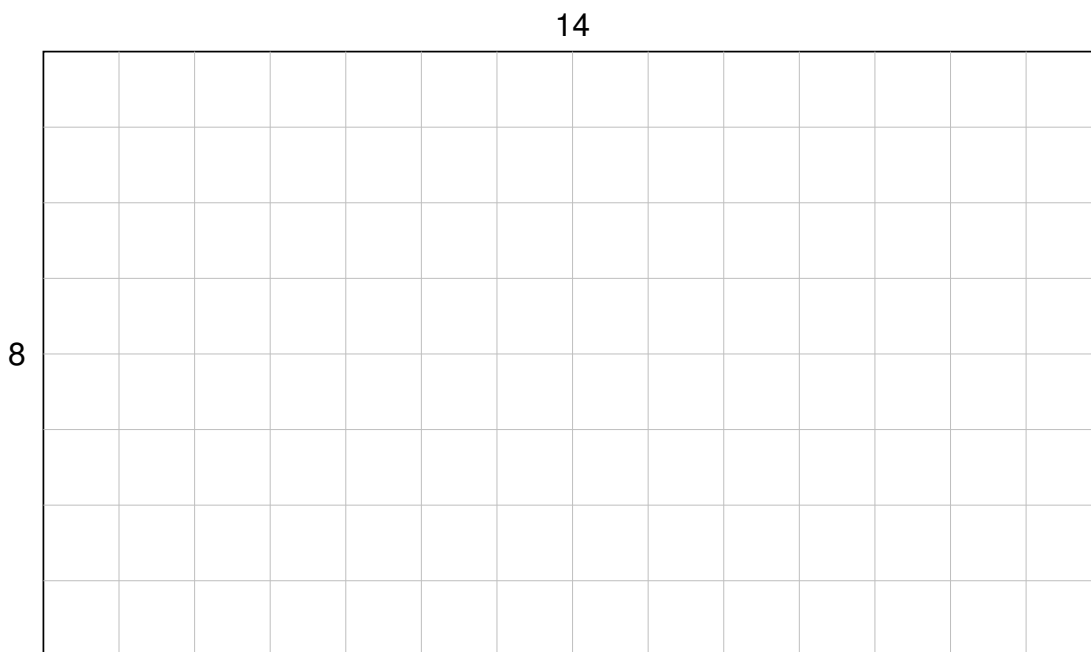
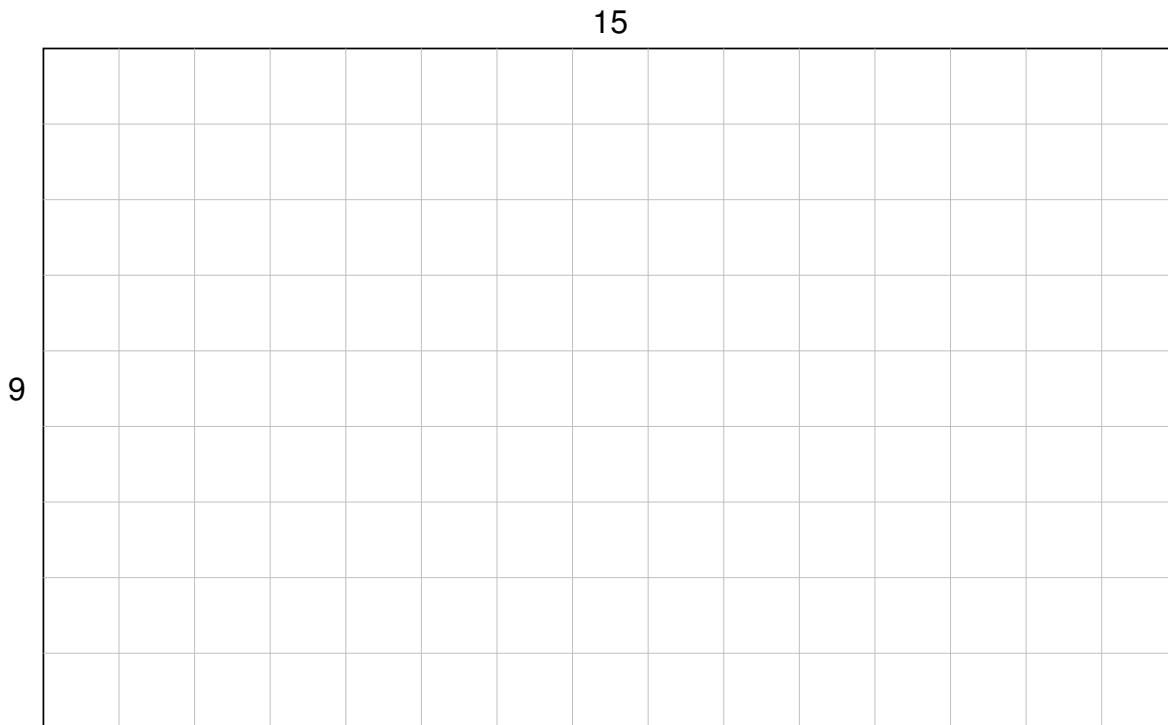


Problem 1: Apply the biggest-square rule to both rectangles. Record every square size, including repeated sizes, and compare the last squares.

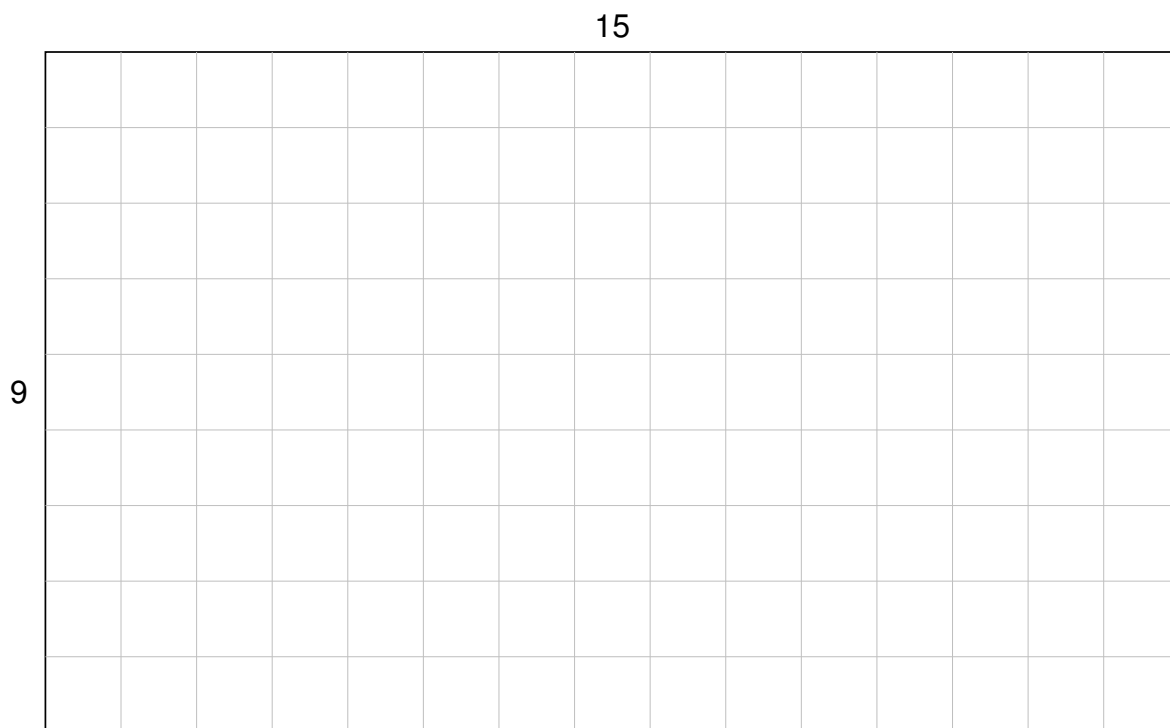
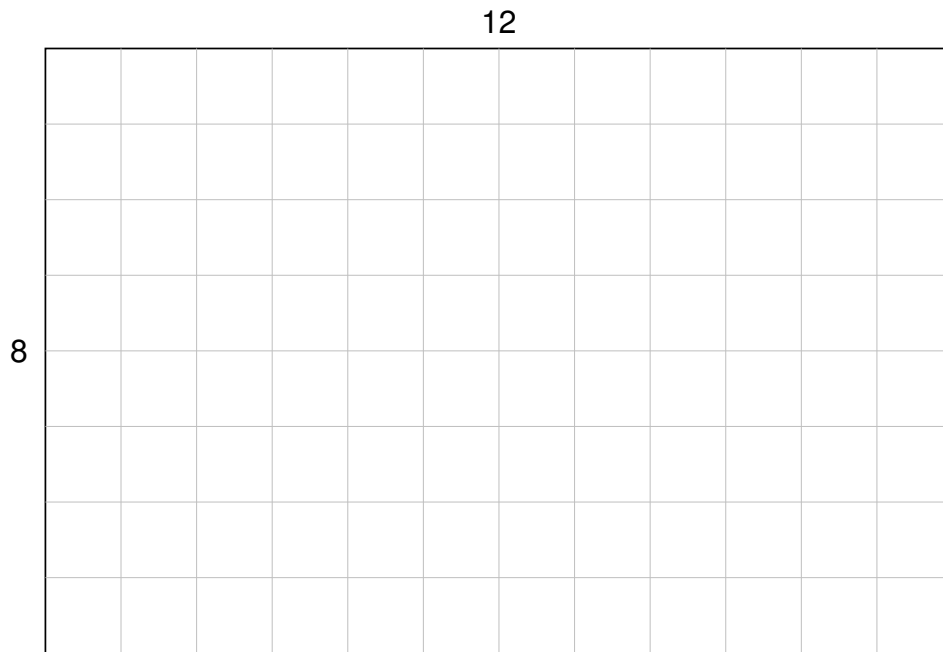




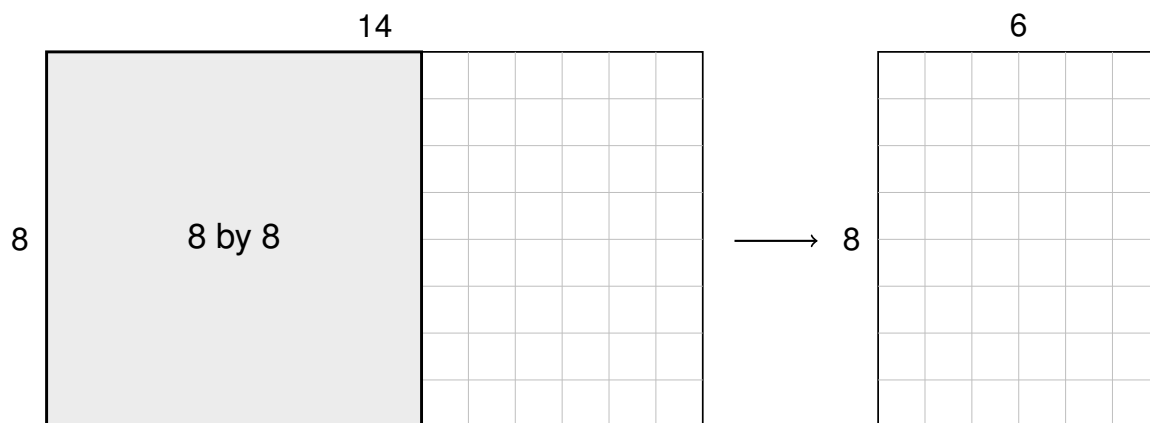
Problem 2: Predict the last square in each rectangle, then test. Find a rule that predicts the last square from the two starting side lengths.



Problem 3: For each rectangle, find every size of identical whole-grid square that can cover it. Explain why your list is complete.



Problem 4: A 14 by 8 rectangle loses an 8 by 8 square and leaves a 6 by 8 rectangle. Which whole-number lengths measure both edges exactly before the cut? Which work afterward? Explain why these two lists always match after a biggest-square cut.



Problem 5: Explain why the last square from the biggest-square rule is the largest whole-grid square that can tile the starting rectangle using identical copies.

Problem 6: Choose a longer side from 9 through 15 and a shorter side from 2 through 8. Which rectangle produces the most different square sizes under the biggest-square rule? Find every rectangle that ties.

